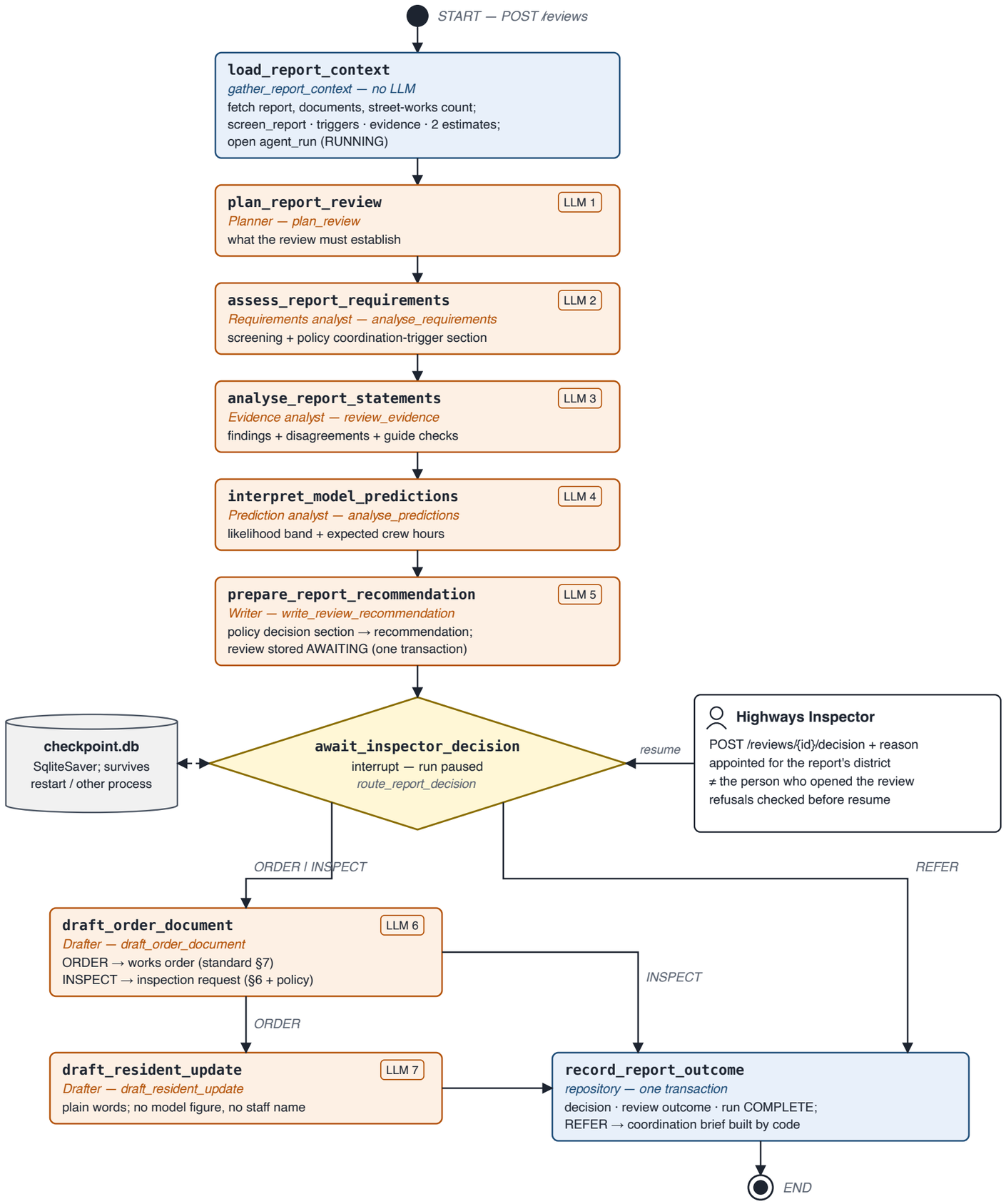
StreetCare AI — report review workflow (LangGraph state graph)

Ten nodes, two conditional branches (both chosen by route_report_decision), one interrupt. Nodes return only the keys they write.



Design invariants

Facts before prose	every figure, result and passage an agent sees was computed by tools/ (standards, register, models, documents); agents compute nothing
One call per agent	prompt → labelled fields → parse_labelled_response; each call is recorded in llm_calls (tokens, latency) as soon as the agent returns
Stop before deciding	nothing is decided, written or sent on a model's word
Four-eyes rule	decider ≠ opener; decider appointed for the report's district
Durable pause	interrupt state persisted; resumable by another process after restart
Re-entrant gate	a paused node re-runs from its start on resume, so it writes nothing
Bounded cost	5 LLM calls before the gate; at most 7 per run

ReportReviewState (TypedDict, agentic/state.py)

identity	`report_reference · coordinator_request` `coordinator_id · thread_id · run_id · review_id`

facts (code)	`report_profile · report_documents`
	`register_count · screening · evidence_findings`
	`disagreements · likelihood · crew_hours`

agent replies	`review_plan · requirements_analysis` `evidence_review · prediction_analysis`
decision	`review_decision {decision, decided_by, …}`
outputs	`order_document · document_details` `resident_update`
accounting	`llm_call_count (additive reducer)`

Decision outcomes

Decision	Path after the gate	Written	LLM calls
ORDER	order doc → resident update → record	works order + resident update	5 + 2 = 7
INSPECT	order doc → record	inspection request	5 + 1 = 6
REFER	record (brief by code)	coordination brief	5 + 0 = 5

Review status then follows REVIEW_STATUS_BY_DECISION.